

ASIKKALA PULKKILANHARJU, Finland

WMO: 027270

Lat: 61.2653N

Lon: 25.5203E

Elev: 76

StdP: 100.42

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.1	-21.7	-27.9	0.3	-25.0	-24.1	0.4	-21.7	13.3	-3.7	12.3	-4.2	1.4	330	0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.1	25.6	19.1	23.9	18.1	22.2	17.2	20.4	23.9	19.3	22.8	18.0	21.5	3.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.9	22.6	17.9	12.9	21.7	16.5	11.8	19.9	58.9	24.0	55.3	22.7	51.2	21.7	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	9.6	8.4	-26.1	27.2	6.1	2.0	-30.5	28.7	-34.0	29.9	-37.5	31.0	-41.9	32.5		
			-25.9	21.0	6.3	1.5	-30.5	22.0	-34.1	22.9	-37.7	23.7	-42.3	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-7.4	-5.8	-2.5	3.2	10.4	14.6	17.8	16.2	11.7	5.7	1.6	-2.5
	DBStd	9.56	6.98	7.04	4.84	3.45	4.01	3.31	3.23	2.64	2.92	3.20	3.88	5.50
	HDD10.0	2421	541	443	388	206	44	3	0	0	17	138	253	388
	HDD18.3	4808	799	676	647	455	246	120	50	76	198	393	503	646
	CDD10.0	707	0	0	0	1	57	141	242	193	69	3	0	0
	CDD18.3	52	0	0	0	0	1	7	34	11	0	0	0	0
	CDH23.3	198	0	0	0	0	4	26	141	28	0	0	0	0
	CDH26.7	17	0	0	0	0	0	1	14	2	0	0	0	0
Wind	WSAvg	3.5	3.5	3.7	3.8	3.1	3.2	3.4	3.1	2.8	3.3	3.8	4.1	4.3
Precipitation	PrecAvg	600	48	33	30	27	40	63	84	66	46	66	51	45
	PrecMax	730	91	72	49	55	85	102	167	120	97	153	112	81
	PrecMin	527	12	3	4	6	10	25	16	8	15	12	5	10
	PrecStd	51	19	19	12	14	18	23	40	31	24	33	27	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.9	5.7	8.8	16.9	23.8	25.9	28.2	26.5	20.1	13.6	10.4	8.3
		MCWB	3.0	3.1	4.6	9.7	14.8	17.5	19.7	19.6	16.3	11.8	9.1	7.5
	2%	DB	2.7	3.6	6.1	13.1	21.0	23.8	26.5	23.8	18.2	12.1	8.9	5.4
		MCWB	2.0	2.5	3.1	7.6	14.0	16.7	20.0	18.2	15.1	10.7	8.2	4.5
	5%	DB	1.7	2.3	4.3	10.4	19.0	21.9	24.9	22.1	16.9	11.3	7.5	4.2
		MCWB	1.2	1.5	2.1	6.0	13.1	15.8	19.3	17.1	14.3	10.1	6.8	3.4
	10%	DB	0.7	1.3	3.0	8.1	16.9	20.0	23.1	20.6	15.8	10.2	6.3	3.1
		MCWB	0.3	0.9	1.5	4.7	11.5	15.1	18.5	16.4	13.8	9.2	5.7	2.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	3.5	5.2	10.1	16.4	20.1	21.6	21.1	16.9	12.2	9.2	7.6
		MCDB	3.7	4.7	7.3	15.3	19.9	22.8	25.6	24.7	18.7	13.2	10.1	8.3
	2%	WB	2.2	2.5	3.6	8.2	15.1	17.9	20.8	19.3	15.8	11.2	8.2	4.7
		MCDB	2.8	3.4	5.5	12.0	19.3	21.9	24.6	22.5	17.3	11.9	9.0	5.2
	5%	WB	1.2	1.7	2.6	6.5	13.9	16.7	20.1	18.0	15.0	10.4	7.0	3.5
		MCDB	1.6	2.3	3.9	9.9	17.8	20.5	23.7	20.9	16.5	11.1	7.6	4.1
	10%	WB	0.4	0.9	1.7	5.1	12.5	15.7	19.1	17.0	14.1	9.2	5.8	2.4
		MCDB	0.8	1.3	2.7	7.4	16.5	19.2	22.7	19.5	15.4	10.1	6.2	2.9
Mean Daily Temperature Range	5% DB	MDBR	5.5	5.3	6.7	7.2	7.6	6.6	6.1	5.8	4.8	3.9	3.1	3.7
		MCDBR	3.3	4.3	7.2	11.5	10.9	9.4	8.0	7.8	6.3	4.9	4.2	4.1
		MCWBR	3.3	3.8	5.1	6.4	5.5	4.6	4.0	3.9	3.9	4.1	4.0	4.0
	5% WB	MCDBR	3.4	4.1	6.1	10.1	9.5	8.1	7.3	6.8	5.8	4.8	4.0	4.1
		MCWBR	3.3	3.7	4.6	6.4	5.4	4.5	4.0	4.0	3.9	4.4	3.9	4.0
Clear-Sky Solar Irradiance	taub	0.255	0.298	0.325	0.363	0.360	0.356	0.379	0.366	0.341	0.323	0.280	0.364	
	taud	2.082	2.152	2.229	2.342	2.419	2.459	2.410	2.439	2.500	2.404	2.177	2.595	
	Ebn at Noon	452	663	784	828	863	873	838	820	778	649	422	303	
	Edh at Noon	39	71	93	102	102	100	102	92	73	56	36	25	
All-Sky Solar Radiation	RadAvg	0.23	0.80	2.27	3.72	5.02	5.26	5.09	3.94	2.32	0.99	0.27	0.13	
	RadStd	0.04	0.18	0.29	0.39	0.56	0.41	0.50	0.45	0.31	0.17	0.05	0.02	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (54 neighbors)	+0.48	N/A	N/A	N/A	N/A	N/A	-146	N/A	N/A	N/A	

Nomenclature: See separate page